

Practice problems #18

1. We know that $\mathbf{R}^{2 \times 2}$ is a vector space with usual element-wise addition and scalar multiplication.

- (a) Show that the set H of all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ is a subspace of $\mathbf{R}^{2 \times 2}$.
- (b) Give a spanning set for H .

2. Compute the determinant of the matrix A using

- (a) cofactor expansion
- (b) short-cut formula given in class.

$$A = \begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \\ 2 & -4 & 3 \end{bmatrix}.$$

3. Compute the determinant.

$$A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 2 & 7 & 0 & 6 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -5 \end{bmatrix}$$

Answers or Hints on page 2.

#1(a): Check 4 items as we did in class.

#1(b): $H = \text{span}$ of what *vectors*? (Here, vectors are matrices.) One *possible* answer is

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

#2: 0 for both

#3: -546